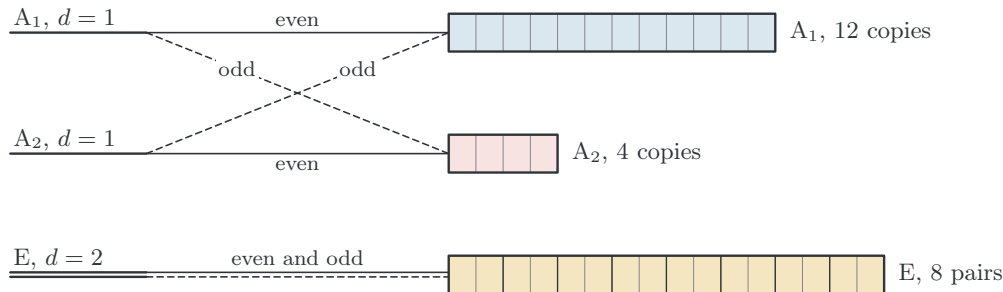


$\mathcal{H}_{\text{spin}} = (\mathbb{C}^2)^{\otimes 5}$ for the five protons, $\dim = 32$; under G_6 it holds 12 copies of A_1 , 4 of A_2 and 8 of E, so $12 + 4 + 2 \cdot 8 = 32$; at right each spin species is a bar of one unit per dimension

spatial species Γ , dimension d_Γ

spin species, copies in $\mathcal{H}_{\text{spin}}$



the weight of a spatial level is the length of the bar it is joined to: $w_{A_1}(+) = 12$, $w_{A_1}(-) = 4$; $w_{A_2}(+) = 4$, $w_{A_2}(-) = 12$; $w_E(\pm) = 8$ for each member; $\sum_\Gamma d_\Gamma w_\Gamma(\pm) = 32$

rule: $\Gamma_{\text{space}} \otimes \Gamma_{\text{spin}}$ must contain the parity character, $\chi_+ = A_1$ (solid) or $\chi_- = A_2$ with $b \mapsto -1$ (dashed); $\chi_{\text{stat}} \equiv +1$ on G_6 , every permutation there being even; A_2 has no copy among the three version states (fig. 7) and is still a valid spin sector

level rules and connections in the manner of Harter (1993) fig. 4.2.3; a double rule marks the two-dimensional E